Problem Set VI: Edgeworth Box

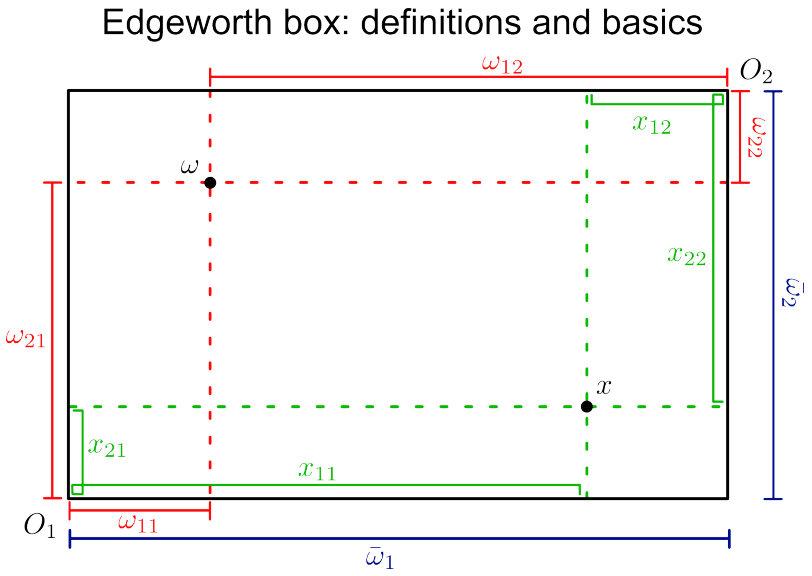
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Exercises solved in class on *March 15th, 2010*

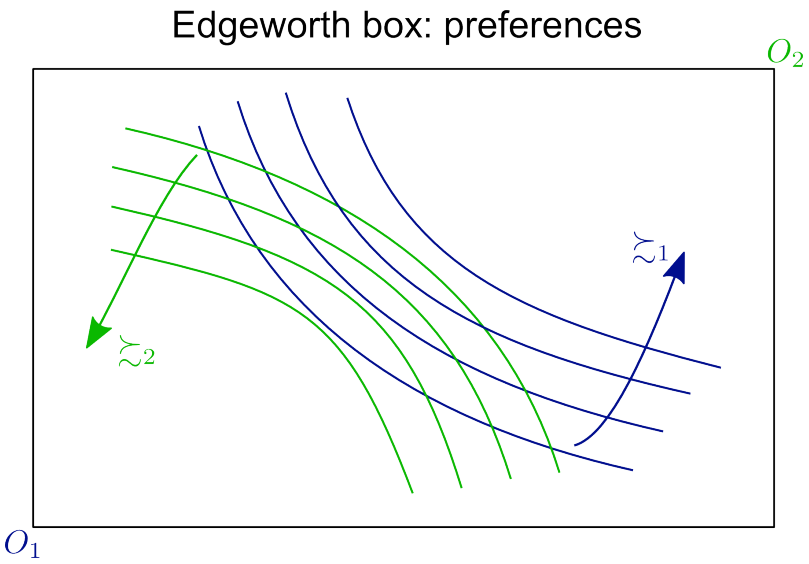
Recap: pure exchange

- The simplest model of a general equilibrium theory is a model with no production: a pure exchange economy.
- Consumers have endowments $\omega = ((\omega_{11}, \omega_{21}), (\omega_{12}, \omega_{22}))$, and locally non-satiated preferences; they can exchange goods in order to increase their level of utility.
- An allocation $x = ((x_{11}, x_{21}), (x_{12}, x_{22}))$ represents the amounts of each good that are allocated to each consumer.
- A nonwasteful allocation is one for which $x_{l1} + x_{l2} = \bar{\omega}_l$, for $l = 1, 2$.
- Given two consumers, two goods, and no production, all non-wasteful allocations can be drawn in an Edgeworth Box.
- Every point in the box represents a complete allocation of the two goods to the two consumers.
- We will analyse the exchanges in the Edgeworth Box, to find an equilibrium outcome.

Recap: Edgeworth Box basics



Recap: Edgeworth Box preferences



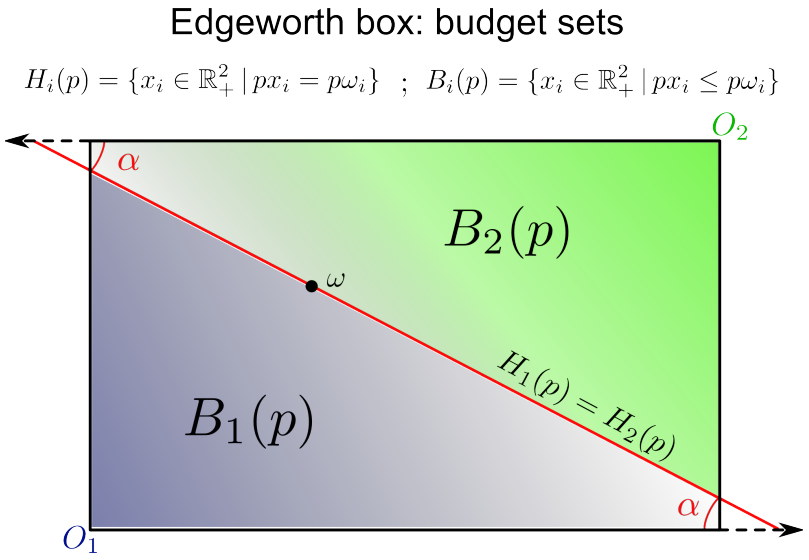
Main concepts of an EB

- The wealth of the consumers is not given exogenously: it is determined by the value of their endowment at the prices that will prevail in the exchange process.
- Hence, the budget set of each consumer is given by:

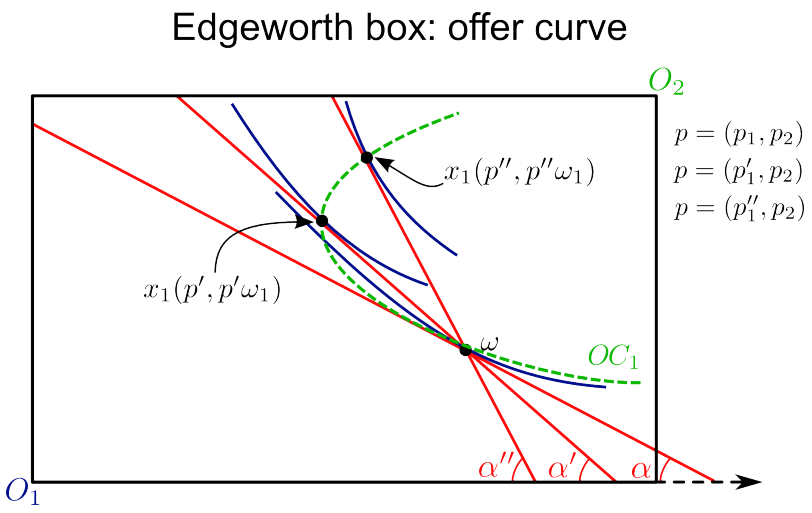
$$B_i(p) = \{x_i \in \mathbb{R}^2_+ : \ p \cdot x_i \leq p \cdot \omega_i\}$$

- for every level of prices, consumers will face a different budget set.
- the locus of preferred allocations for every level of prices is the consumer's *Offer Curve*.

Recap: Edgeworth box budget set



Recap: Edgeworth box offer curve OC



Efficiency and equilibrium

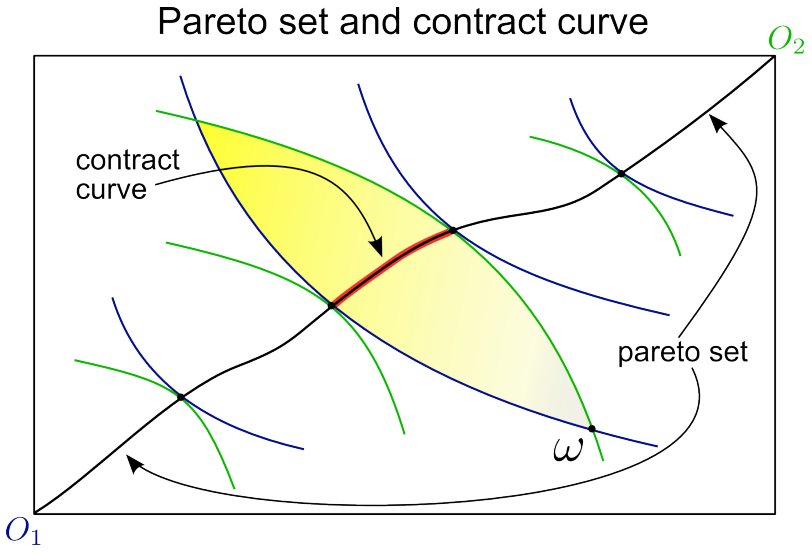
- Given the setting described above, we are now interested in the *efficiency* of exchange.
- An allocation is said to be *Pareto efficient*, or *Pareto Optimal*, if there is no other feasible allocation in the economy for which both are at least as well off and one is strictly better off.
- formally,

$$x \text{ is P.O. if } \nexists\, x' \text{ s.t. } x_i' \succsim_i x_i, \ \forall\, i, \text{ and } \exists i \text{ s.t. } x_i' \succ_i x_i$$

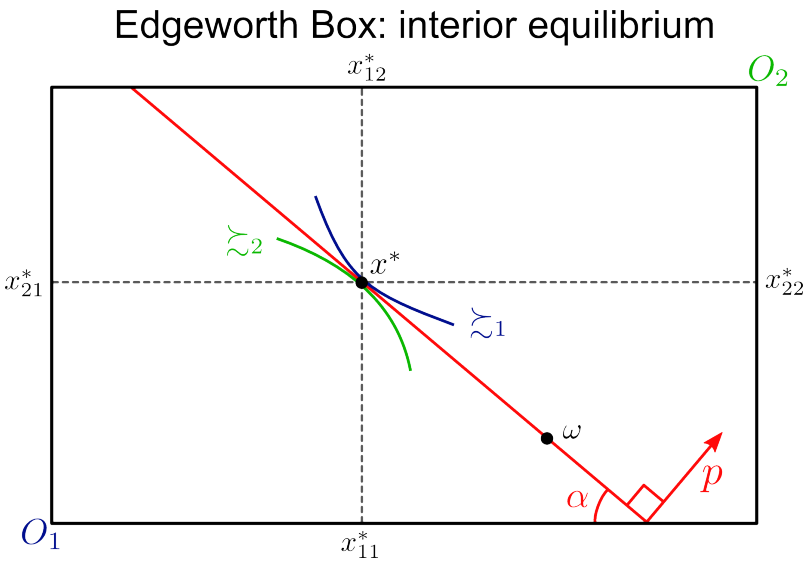
- The locus of points that are P.O. given preferences and endowments is the *Pareto Set*.
- The part of the Pareto Set in which both consumers do at least as well as their initial endowments is the *Contract Curve*.
- Moreover, we are interested in the equilibrium point(s) of the process of exchange:
- a Walrasian equilibrium is a price vector p^* and an allocation x^* such that, for every consumer

$$x_i^* \succsim_i x_i' \text{ for all } x_i' \in B_i(p^*)$$

Recap: contract curve, pareto set



Recap: Edgeworth Box equilibrium

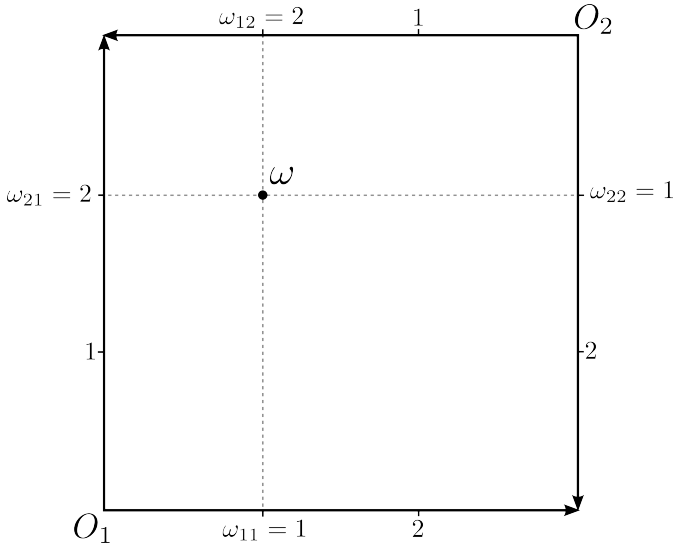


1. Edgeworth box

Consider a pure-exchange, private-ownership economy, consisting in two consumers, denoted by $i = 1, 2$, who trade two commodities, denoted by $l = 1, 2$. Each consumer i is characterized by an endowment vector, $\omega_i \in \mathbb{R}^2_+$, a consumption set, $X_i = \mathbb{R}^2_+$, and regular and continuous preferences, $\succsim_i$ on X_i.

1. Assuming that the consumers' endowments are $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$, respectively, construct the Edgeworth Box relative to economy under consideration. With reference to the same economy, define the following notions: competitive equilibrium, Pareto-efficient allocation, Pareto set, contract curve.
2. Find the equation describing the Pareto set (internal solutions); then, taking commodity 1 as the numeraire, hence positing $p_1 \equiv 1$, find the competitive equilibrium allocation and price system; and, finally, draw your results in the Edgeworth Box in each of the following two cases:
   (a) both consumers' preferences are represented by the same Cobb-Douglas utility function: $u_i(x_{1i}, x_{2i}) = x_{1i}^{\frac{1}{3}} x_{2i}^{\frac{2}{3}}$;
   (b) the two consumers' preferences are respectively represented by the following quasilinear utility functions: $u_1(x_{11}, x_{21}) = x_{11} + \ln x_{21}$; $u_2(x_{12}, x_{22}) = x_{12} + 2 \ln x_{22}$.
3. Explain in which of the above two cases the preferences are homothetic and in which case, instead, the preferences are such as to rule out the wealth effects usually associated with price changes. How such peculiar properties of consumers' preferences affect the Pareto set and the shadow prices, i.e., the prices implicit in the Pareto-efficient allocations?

Cobb-Douglas: Edgeworth Box



Cobb-Douglas: Finding Pareto Set

- The Pareto Set is the set of allocations that are P.O.
- To find the equation of the interior Pareto Set we must find the locus of points for which the marginal rates of substitution of the two consumers match.
- Only at those points, in fact, both consumers will be 'happy', not willing to change their behaviour.

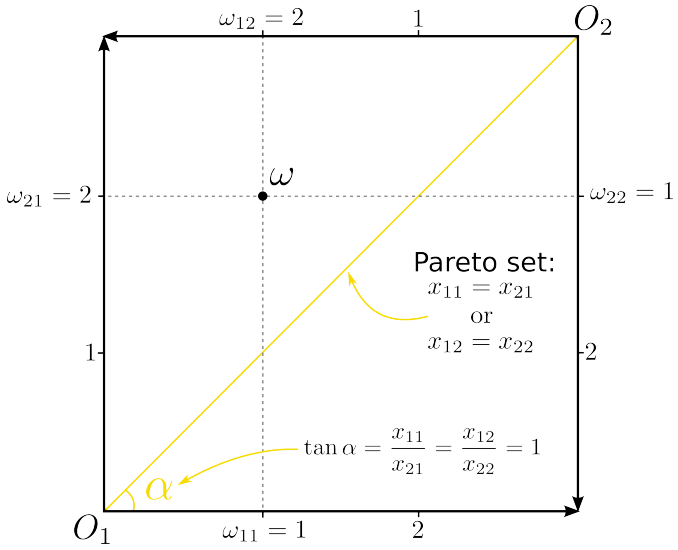
$$MRS^1_{12} = \frac{\partial u_1(x_{11}, x_{21})}{\partial x_{11}} \left(\frac{\partial u_1(x_{11}, x_{21})}{\partial x_{21}} \right)^{-1} = \frac{1}{2} \frac{x_{21}}{x_{11}}$$

$$MRS^2_{12} = \frac{\partial u_2(x_{12}, x_{22})}{\partial x_{12}} \left(\frac{\partial u_2(x_{12}, x_{22})}{\partial x_{22}} \right)^{-1} = \frac{1}{2} \frac{x_{22}}{x_{12}} = \frac{1}{2} \frac{\bar{\omega}_2 - x_{21}}{\bar{\omega}_1 - x_{11}} = \frac{1}{2} \frac{3 - x_{21}}{3 - x_{11}}$$

$$MRS^1_{12} = MRS^2_{12} \Rightarrow \frac{1}{2} \frac{x_{21}}{x_{11}} = \frac{1}{2} \frac{3 - x_{21}}{3 - x_{11}}$$

Pareto set equation: $x_{21} = x_{11}$, i.e. $x_{22} = x_{12}$

Cobb-Douglas: Pareto set



Cobb-Douglas: Finding Contract curve

- The contract curve is the portion of the pareto set for which consumers are at least as well off as by staying with their endowment.
- We have hence to put into a system the indifference curve passing by the endowment point and the pareto set.
- By definition, a curve of indifference is the locus of points with the same level of utility
- we will hence calculate the level of utility at the endowment allocation, and combine it with the pareto set equation.

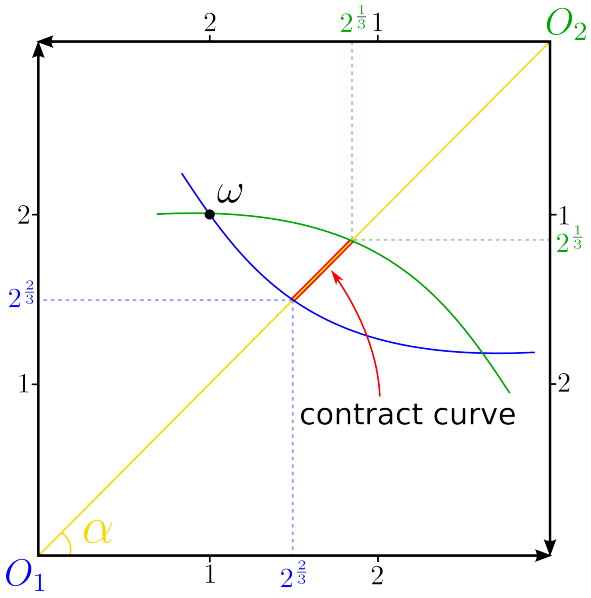
$$U_1(\omega_{11}, \omega_{21}) = (1)^{\frac{1}{3}}(2)^{\frac{2}{3}} = 2^{\frac{2}{3}} \;\; ; \;\; U_1(\omega_{11}, \omega_{21}) = (2)^{\frac{1}{3}}(1)^{\frac{2}{3}} = 2^{\frac{1}{3}}$$

$$\text{For } i = 1, \begin{cases} x_{11} = x_{21} \\ x_{11}^{\frac{1}{3}} x_{21}^{\frac{2}{3}} = 2^{\frac{2}{3}} \end{cases} \;\; ; \;\; \text{For } i = 2, \begin{cases} x_{21} = x_{22} \\ x_{21}^{\frac{1}{3}} x_{22}^{\frac{2}{3}} = 2^{\frac{1}{3}} \end{cases}$$

Which can be solved to give the extremal points of the contract curve

$$x_{11} = x_{21} = 2^{\frac{2}{3}} \;\text{ and }\; x_{21} = x_{22} = 2^{\frac{1}{3}}$$

Cobb-Douglas: Contract curve



Cobb-Douglas: Finding equilibrium, I

- Let's impose $p_1 \equiv 1$ and $p_2 \equiv p$.
- given these prices and endowments, wealth for the two consumer is:

$$w_1 = p_1\omega_{11} + p_2\omega_{21} = 1 + 2p \;\; ; \;\; w_2 = p_1\omega_{21} + p_2\omega_{22} = 2 + p$$

- Each consumer solves a maximisation problem given his budget constraint;
- From the maximisation we get each consumer demand function for each good;
- Then, by using the fact that $x_{l1} + x_{l2} = \bar{\omega}_l, \; \forall \; l = 1, 2$, we can close the system, finding:
  1. the vector of equilibrium prices $p^* = (1, p_2^*)$;
  2. the equilibrium allocation x^*.

The max problem of player 1 is

$$\max \;\; x_{11}^{\frac{1}{3}} x_{21}^{\frac{2}{3}}, \;\; \text{s.t.:} \;\; x_{11} + p x_{21} = 1 + 2p$$

To reuse previous calculations, remember $MRS_{12}^i = \dfrac{p_1}{p_2}$.

Cobb-Douglas: Finding equilibrium, II

- The solution of the two maximisation problems gives us the four demand functions:

$$x_{11}(1, p) = \frac{1}{3}(1 + 2p) \;\; ; \;\; x_{21}(1, p) = \frac{2}{3}\left(\frac{1 + 2p}{p}\right)$$

$$x_{12}(1, p) = \frac{1}{3}(2 + p) \;\; ; \;\; x_{22}(1, p) = \frac{2}{3}\left(\frac{2 + p}{p}\right)$$

- These demand functions can be summed over each good to get the total demand for each good;
- Then, by imposing $x_{l1} + x_{l2} = \bar{\omega}_l, \; \forall \; l = 1, 2$ we find prices p^*:

$$x_1 = x_{11}(1, p) + x_{12}(1, p) = 1 + p \;\; ; \;\; x_2 = x_{21}(1, p) + x_{22}(1, p) = 2\frac{1 + p}{p}$$

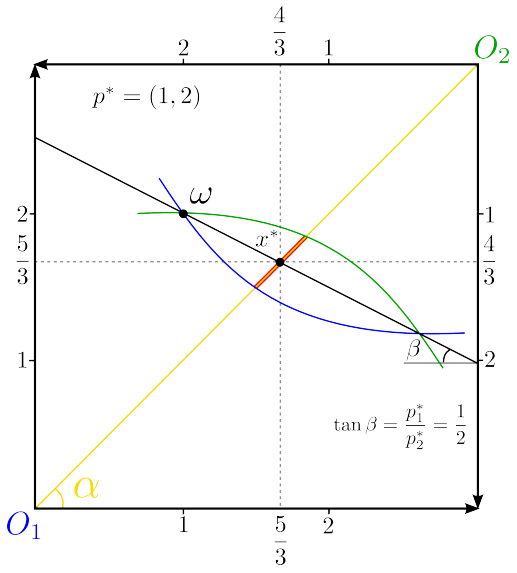
Then, by imposing the condition we get

$$x_1 = 3 \;\Rightarrow\; 1 + p = 3 \;\Rightarrow\; p^* = 2 \;,\; x_2 = 3 \;\Rightarrow\; 2\frac{1+p}{p} = 3 \;\Rightarrow\; p^* = 2$$

By plugging this into the demand of each consumer, we get

$$x_{11}^* = x_{21}^* = \frac{5}{3} \;,\; x_{21}^* = x_{22}^* = \frac{4}{3}$$

Cobb-Douglas: Equilibrium



Quasilinear: results

- The solution strategy for quasilinear preferences is the same as the strategy for the Cobb-Douglas case.
- We will not show all of the calculations, but just results and a graph.
- Do it yourself: good luck!

Pareto set: $x_{21} = 1$

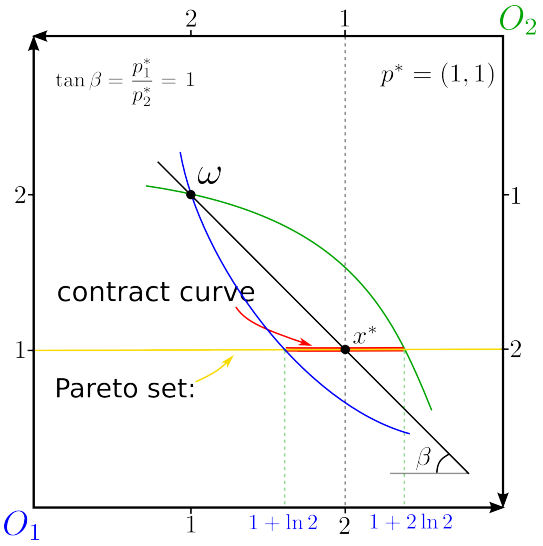
Contract curve: $x \in$ Pareto Set s.t. $1 + \ln 2 \leq x_{11} \leq 1 + 2\ln 2$

Demand x_2**:** $x_{21}(p, \omega) = \frac{p_1}{p_2}$; $x_{22}(p, \omega) = \frac{2p_1}{p_2}$

Eq prices: $p^* = (1, 1)$

Eq allocation: $x^* = ((2, 1), (1, 2))$

Quasilinear: graphics



Cobb Douglas: homothetic

- A preference relation is homothetic if indifference sets are related by proportional expansion along rays.
- Take the $MRS_{12}^1 = \frac{1}{2}\frac{x_{21}}{x_{11}}$; take a $\alpha > 0$
- Take a ray starting from the origin, equation $x_{21} = \alpha x_{11}$.
- Then $\frac{1}{2}\frac{\alpha x_{11}}{x_{11}} = \frac{\alpha}{2}$ constant: preferences are homothetic.
- Homotheticity implies that a ray through the origin is also a isocline.
- Since we have seen that in this case, with identical preferences for the two consumers, the Pareto set is the diagonal of the Box, it is a ray through the origin for both players, and hence it is an isocline.
- The relative shadow prices are constant, at $\frac{1}{2}$

Quasilinear: no wealth effects

- The quasilinear preferences, instead, rule out wealth effects.
- This implies that the contrat curve is horizontal...
- ...since there is no wealth effect for good 2.
- To check this, just have a look at the Walrasian demand for good two: it does not depend on wealth.
- Indifferent curves are parallel.
- Shadow prices implicit in the Pareto Set are constant (equal to one in this case).

2. MWG 15.B.1 + 15.B.3: Edgeworth box

Consider an Edgeworth box economy with two goods and two consumers with locally non satiated preferences. Let $x_{li}(p)$ be consumer i's demand of good l at prices $p = (p_1, p_2)$.

1. Show that $p_1(\sum_i x_{1i}(p) - \bar{\omega}_1) + p_2(\sum_i x_{2i}(p) - \bar{\omega}_2) = 0$ for all prices p.
2. Argue that if the market for good 1 clears at prices $p^* \gg 0$, then so does the market for good 2; hence, p^* is a Walrasian equilibrium price vector.
3. Argue graphically that the Walrasian equilibrium is Pareto optimal.

Point 1, I

Solution strategy
The relation just tells us that demand and supply of goods cancel out at any price level.

- We will first show that maximising behaviour implies that consumer demand must sit at the boundary of the budget set;
- And then, we will use the budget set itself to find the required expression

Let's first assume that the budget set holds with strict inequality:

$$px_i < p\omega_i \text{ i.e.: } \quad p_1x_{1i}(p) + p_2x_{2i}(p) < p_1\omega_1 p_2\omega_2$$

...but since demand $x_{li}(p)$ is locally non-satiated, there must exist a bundle that is strictly preferred to the above bundle:

$$\exists\,(x_{1i}, x_{2i}) \succ_i (x_{1i}(p), x_{2i}(p))$$

This will be true until we reach the boundary of the budget set, that will hence hold with equality.

Point 1, II

Since the budget constraint holds with equality, we can write

$$p_1x_{1i}(p) + p_2x_{2i}(p) = p_1\omega_1 + p_2\omega_2$$

That is equivalent, collecting by p_1 and p_2, to

$$p_1(x_{1i}(p) - \omega_1) + p_2(x_{2i}(p) - \omega_2) = 0$$

It is now possible to sum over i to get the aggregated result, that is

$$p_1(\sum_i x_{1i}(p) - \bar{\omega}_1) + p_2(\sum_i x_{2i}(p) - \bar{\omega}_2) = 0$$

Market clearing

If the market for good one clears, it means that all the available quantity of good 1, $\bar{\omega}_1$, has been demanded by one or the other trader. This means that

$$\sum_i x_{1i}(p^*) - \bar{\omega}_1 = 0$$

By using the above result, $p_1(\sum_i x_{1i}(p) - \bar{\omega}_1) + p_2(\sum_i x_{2i}(p) - \bar{\omega}_2) = 0$, since the first element is zero, the second too must be zero, and hence

$$\sum_i x_{2i}(p^*) - \bar{\omega}_2 = 0$$

So also the second market clears.

Pareto Optimum, Definition, intuition

Definition 1 (Pareto Optimum)**.** An allocation x in the Edgeworth box is Pareto optimal if there is no allocation x' with $x'_i \succsim_i x_i$ for $i = 1, 2$ and $x'_i \succ x_i$ for some i

- That is, both traders are at least as well off, and at least one of the two is strictly better off.
- Graphically, this means that the upper contour sets of the two consumers do not intersect...
- Because if they did, there would be an allocation x' that would make both of them strictly better off.

Solution strategy
Hence, we will define the upper contour sets for the two consumers in the case of a Walrasian equilibrium, and we will show that they have no intersection point other than the Walrasian equilibrium itself.

Pareto Optimum, analytical argument, I

- Since the consumer is locally non-satiated, at the Walrasian equilibrium (x^*, p^*):
- the upper contour set lies above the budget constraint or on it;
- the strict upper contour set lies strictly above the budget constraint.

For player one, we define the UCS as A and the strict UCS as B; for player two, as C and D. They are:

$$A = \{x_1 \in \mathbb{R}^2_+ : \ x_1 \succsim_1 x_1^*\} \ , \ B = \{x_1 \in \mathbb{R}^2_+ : \ x_1 \succ_1 x_1^*\}$$

$$C = \{x_2 \in \mathbb{R}^2_+ : \ x_2 \succsim_2 x_2^*\} \ , \ D = \{x_2 \in \mathbb{R}^2_+ : \ x_2 \succ_2 x_2^*\}$$

Pareto Optimum, analytical argument, II

- By the definition of P.O., if there exists an x' that is weakly preferred to x by both and strictly preferred by one, then x is not P.O.
- Given the sets described above, we can show that points having the above property do not exist;
- hence, the Walrasian equilibrium x^* must be a Pareto Optimal allocation.

The points that make consumer one at least as well off and consumer two strictly better off are given by $A \cap D$:

$$A \cap D = \{x_1 \in \mathbb{R}^2_+ : \ x_1 \succsim_1 x_1^*\} \cap \{x_2 \in \mathbb{R}^2_+ : \ x_2 \succ_2 x_2^*\} = \varnothing$$

Similarly, the points that are indifferent for consumer two and strictly preferred by consumer one are $B \cap D$:

$$B \cap C = \{x_1 \in \mathbb{R}^2_+ : \ x_1 \succ_1 x_1^*\} \cap \{x_2 \in \mathbb{R}^2_+ : \ x_2 \succsim_2 x_2^*\} = \varnothing$$

Hence, the allocation x^* is Pareto Optimal.

Pareto Optimum, Graphics

